

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
B.Tech Degree S6 (S, FE) Examination December 2024 (2019 Scheme)

Course Code: ADT302

Course Name: CONCEPTS IN BIG DATA ANALYTICS

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- | | | |
|----|---|-----|
| 1 | List out the 5 Vs of Big Data. | (3) |
| 2 | Write a note on any three big data analytics tools. | (3) |
| 3 | How does stream processing benefit over traditional data processing? | (3) |
| 4 | Some websites check availability of username by searching millions of usernames registered with it. Identify one effective method to filter data as in this type of scenario. | (3) |
| 5 | Explain the functions of data node and name node. | (3) |
| 6 | What are the uses of Apache Hadoop? | (3) |
| 7 | Identify the ways in which a pig program can be executed. | (3) |
| 8 | Write note on Hive Metastore? | (3) |
| 9 | List out the key features of R Programming Language. | (3) |
| 10 | Write an R program to find sum of two numbers. | (3) |

PART B

Answer one question from each module, each carries 14 marks.

Module I

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|----|---|-----|
| 11 | a) Illustrate Big Data Architecture and explain each component in detail. | (7) |
| | b) Differentiate conventional data and big data. | (7) |

OR

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|----|--|-----|
| 12 | a) Recall and explain different phases in big data analytics life cycle? | (7) |
| | b) Compare the types of Big Data with examples. | (7) |

Module II

- | | | |
|----|---|-----|
| 13 | a) Describe Stream data architecture with an explanation and suitable illustration. | (7) |
|----|---|-----|

- b) Illustrate the working of Bloom filter with examples for (7)
i) Inserting an element
ii) Searching an element.

OR

- 14 a) What are the key challenges in Stream Processing? (7)
b) Suppose we have a window of length N (say $N=24$) on a binary system, we want at all times to be able to answer a query of the form “How many 1’s are there in the last K bits?” for $K \leq N$. Suggest an algorithm to solve this issue with detailed explanation. Find the total number of ones, when 0111 enters into the given stream101011000101110110010110.... (Assume, the new bit enters from the right side and time stamp of first new bit is 100).

Module III

- 15 a) Describe Hadoop ecosystem with neat sketch. (7)
b) Explain in detail the anatomy of file read operation in HDFS with neat diagram. (7)

OR

- 16 a) How Can the Word Count Problem Be Implemented Using the MapReduce Model? (14)

Module IV

- 17 a) How Does Pig Compare to Hive? (7)
b) Explain Pig architecture in detail. What are its components? (7)

OR

- 18 a) What is HBase and what are its features? Explain HBase Data Model. (7)
b) What are the different data types used in Pig Latin and describe its various operators. (7)

Module V

- 19 a) Explain in detail different Matrix handling operations used in R Programming Language. (8)
b) List and explain three R functions used in descriptive statistics. (6)

OR

- 20 a) Discuss any four types of data visualization in R. (8)
b) Describe the advantages and disadvantages of data visualization in R language. (6)
